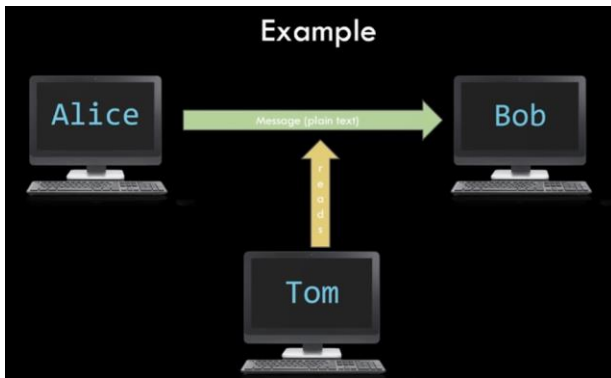


What Is A Network ?

- In simple language a computer network is a collection of computers/devices also known as nodes which are connected to each other in a certain pattern and can communicate with each other using some protocol.
- Protocols are set of rules that define how the communication is going to happen. We will discuss them in future tutorials



What can happen?



When two people communicate over network and transfer data (sending email)(if no security),

Anyone can listen/capture the data transferred (bank account pw...)

INTERRUPTION

An asset of the system is destroyed or becomes unavailable or unusable. It is an attack on **availability**.

Examples:

Destruction of some hardware
Jamming wireless signals
Disabling file management systems

INTERCEPTION

An unauthorized party gains access to an asset. Attack on **confidentiality**.

Examples:

Wire tapping to capture data in a network.
Illicitly copying data or programs
Eavesdropping

No data modification

MODIFICATION

When an unauthorized party gains access and **tamper**s an asset. Attack is on **Integrity**.

Examples:

Changing data file
Altering a program and the contents of a message

FABRICATION

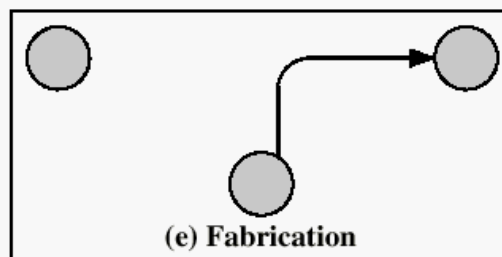
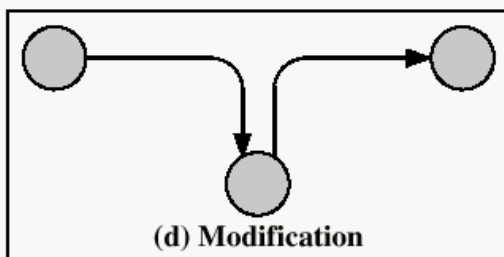
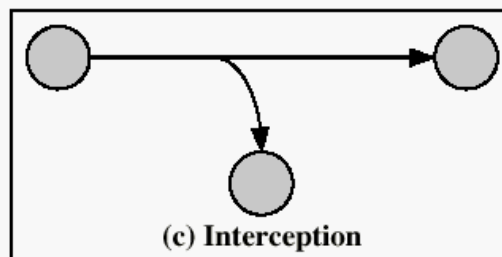
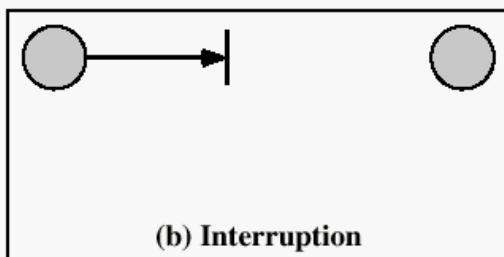
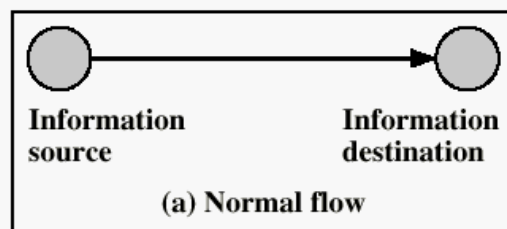
unauthorized party inserts a counterfeit object into the system. Attack on **Authenticity**. Also called impersonation

Examples:

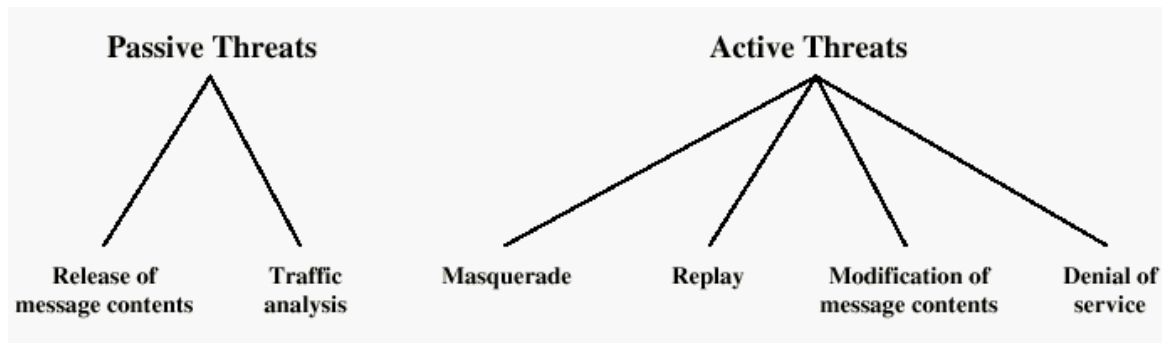
Hackers gaining access to a personal email and sending message

Insertion of records in data files

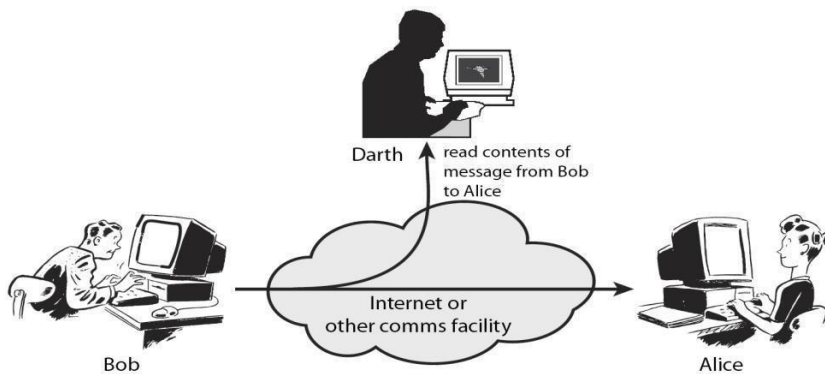
Insertion of spurious messages in a network



Types of Attack



Passive: No data modification. Cannot detect. No harm to resources



Active Attack

Types of Attacks – Active Attacks

- Active attacks are based on **modification** of the **original message** in some manner or the creation of a **false message**.
- These attacks **cannot be prevented easily**. However, they **can be detected** with some effort and attempts can be made to **recover** from them.

Active Attacks - Masquerade

- **Masquerade** is caused when an **unauthorized entity** **pretends** to be another entity.



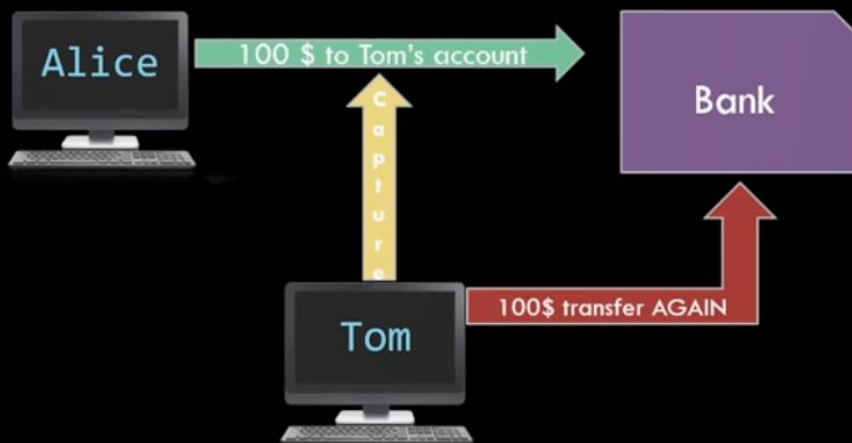
- Lack of authentication

Active Attacks - Modification

- **Modification** results in some alterations to the original message.
- **Modification** results in loss of Integrity
- 2 types:
 - Replay Attacks
 - Alterations

Modification – Replay Attacks

- In replay attacks user captures a sequence of events, or some data units and resends them.

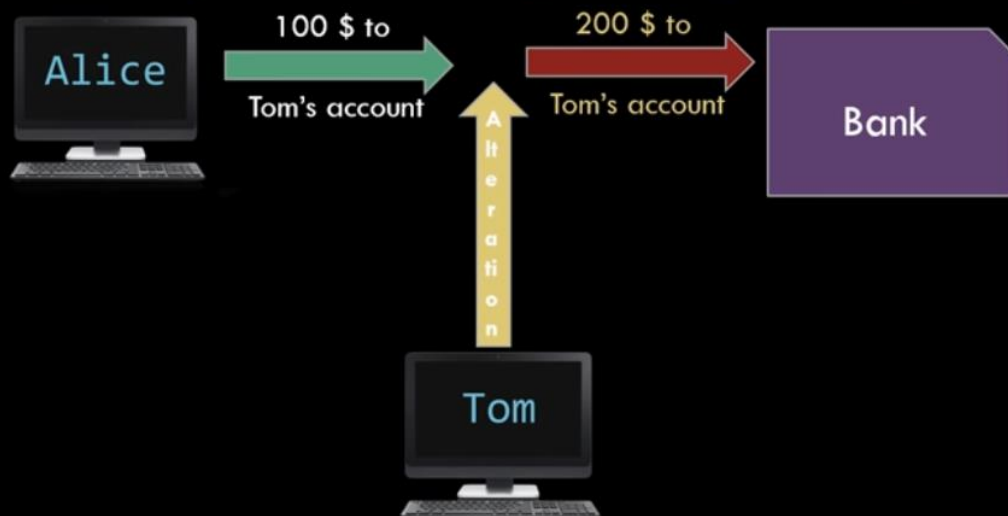


Bank thinks both transactions are from Alice...

In replay attack one can send series of data/message (same msg many times) to receiver so receiver cannot identify which is original message...

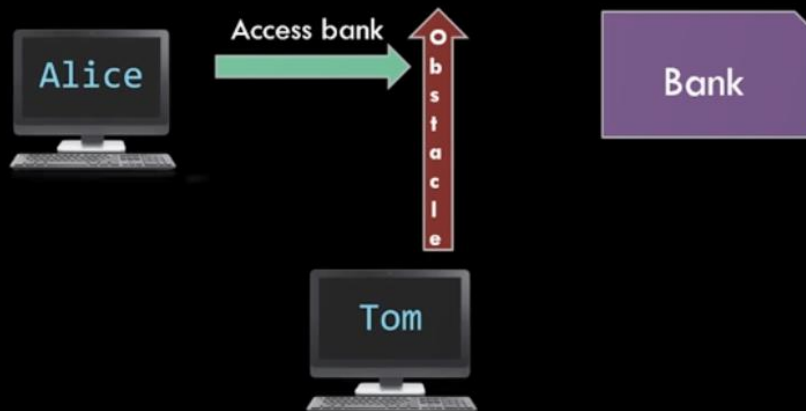
Modification – Alteration

- Alteration of message involves some **change** to the original message.



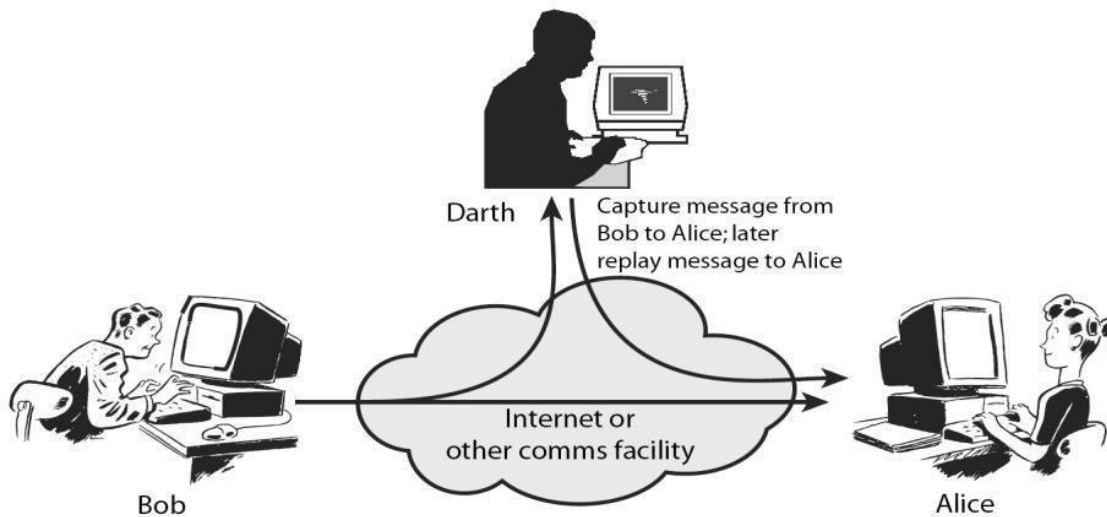
Active Attacks – Denial of Service (DOS)

- Denial of Service (DOS) attacks make an attempt to **prevent** legitimate users from **accessing** some **services**, which they are eligible for.



Alice cannot access bank

Passive Attack

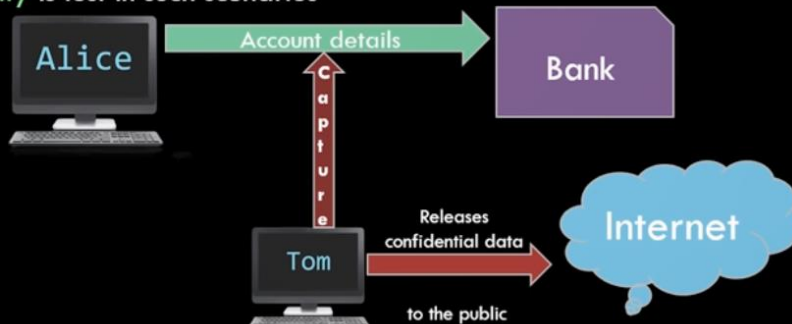


Types of Attacks – Passive Attacks

- **Passive attacks** are those, wherein the attacker indulges in **monitoring of data transmission**. The attacker **aims** to obtain **information** that is in transit.
- The term **passive** indicates that the attacker **does not attempt** to perform any **modifications** to the **data**.
- Passive attacks are **harder to detect**, therefore passive attacks should be prevented rather than detected.
- Passive attacks are further classified into two categories:
 - Release of message contents
 - Traffic analysis.

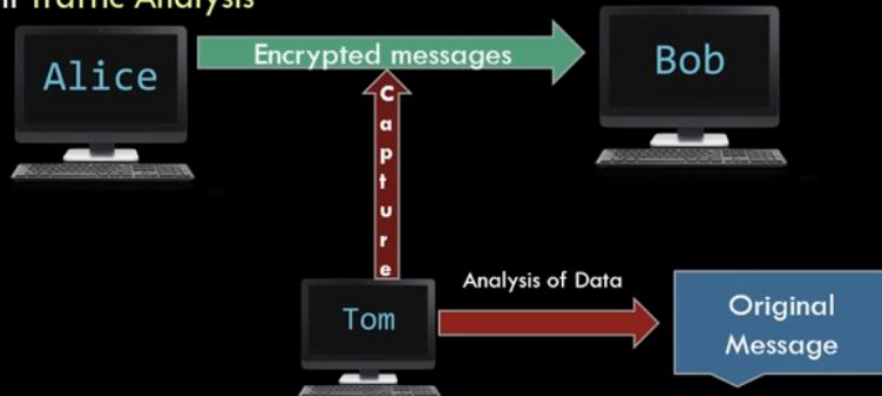
Passive Attacks – Release of Message Content

- In this type of **passive attack** **confidential** data of users is released publicly over the network.
- **Data confidentiality** is lost in such scenarios



Passive Attacks – Traffic Analysis

- In this type of **passive attack** the **attacker** tries to find **similarities** between encrypted messages and finds the original content.
- Such activity is call **Traffic Analysis**



Sr. No.	Key	Active Attack	Passive Attack
1	Modification	In Active Attack, information is modified.	In Passive Attack, information remain unchanged.
2	Dangerous For	Active Attack is dangerous for Integrity as well as Availability.	Passive Attack is dangerous for Confidentiality.
3	Attention	Attention is to be paid on detection.	Attention is to be paid on prevention.
4	Impact on System	In Active Attack, system is damaged.	In Passive Attack, system has no impact.
5	Victim	Victim gets informed in active attack.	Victim does not get informed in passive attack.
6	System Resources	System Resources can be changed in active attack.	System Resources are not changed in passive attack.